

MIE375 — Financial Engineering

Course Notes

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Lecture 1: Introduction/Syllabus

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Principal and Interest

Def. Principal: Amount invested

Interest: "Rent" paid on investment

In general, for

- A : Principal
- r : interest rate per period
- V : Value received.

After 1 period, $V = A(1 + r)$ is received

Interest over multiple periods depend on the compounding rules:

1. **Simple:** Money grows linearly, receiving $V = A(1 + nr)$ over n periods.
2. **Compound:** Money grows geometrically, receiving $V = A(1 + r)^n$ over n periods.

The Rule of 72: Money invested at $r\%$ per year doubles in approximately $72/r$ years.

Ex. Simple: \$5000 borrowed for 5 years at 10% per year simple interest is owed

$$V = A(1 + rn) = 5000(1 + 0.1 \cdot 5) = \$7500$$

Compound: \$100 deposited at 7% compounded yearly for 10 years grows to

$$V = A(1 + r)^n = 100(1 + 0.07)^{10} = \$196.72$$

Simple interest is not used often in practice.

Def. Nominal Interest Rate: annual quote, compounding interest is in fractions corresponding to the compounding period.

For m periods per year for k periods, $V = A(1 + \frac{r}{m})^k$

Ex. Invest \$10 at 8% compounded quarterly for 3 quarters:

$$V = 10 \left(1 + \frac{0.08}{4} \right)^3 = \$10.61$$

Def. Continuous Compound Interest: when the compounding period becomes arbitrarily small, the value approaches e^{rt} .

After 1 year with m periods: $(1 + \frac{r}{m})^m$, and $\lim_{m \rightarrow \infty} (1 + \frac{r}{m})^m = e^r$. More generally, in t years there are $k = tm$ periods, so $(1 + \frac{r}{m})^{tm} \rightarrow e^{rt}$.

Ex. $A = \$100$, $r = 9\%$ compounded continuously. After 1.5 years:

$$V = 100e^{(0.09)(1.5)} = \$114.45$$

Compound Interest Summary

For r the nominal interest rate quoted:

Yearly for k years: $(1 + r)^k$ m periods per year for k periods: $\left(1 + \frac{r}{m}\right)^k$ Continuously for time t: e^{rt}
--

Def. Effective Rate: interest rates are usually quoted as nominal rates, but what you pay depends on the number of compounding periods. The equivalent rate with a *single* compounding period over the time interval of interest is the effective rate.

Relationship between Nominal and Effective Rate (with m compounding periods per year, so the rate per compounding period is $r = r_{nom}/m$):

$$1 + r_{eff} = \left(1 + \frac{r_{nom}}{m}\right)^m \Rightarrow \boxed{r_{eff} = \left(1 + \frac{r_{nom}}{m}\right)^m - 1}$$

1 + Simple One-year Rate = the Explicit Compound

Ex. Effective yearly rate of 8% compounded quarterly:

$$r_{eff} = \left(1 + \frac{0.08}{4}\right)^4 - 1 = 8.24\%$$

Def. APR (Annual Percentage Rate): the *nominal* annual rate of interest including all fees and charges.

APY (Annual Percentage Yield): the *effective* annual rate of interest including all fees and charges.

Both are required by the Truth in Lending Act (1968), and can be used to compare loans with different terms.

Lecture 2: Present Value and Internal Rate of Return

Sep 17, 2026

Present value: comparing cash flows that occur at different times, while interest rates tells how to move cash flows in time. Use interest rates to move all cash flows to the same time, so they become easy to compare.

Ideal Bank: applies the same rate of interest to deposits and loans, and has no transaction costs (real banks have higher rates for loans...). Used for two ways:

- **Compounding:** when cash flow is moved **forward**, **multiplied** by $(1 + r)$ each period.
- **Discounting:** when cash flow is moved **backward**, **divided** by $(1 + r)$ each period.

The Comparison Principle

The ideal bank brings cash flow stream to the same time, then is compared.

- When all cash flows are moved to *present time*, it is called **present value**.
- When all cash flows are moved to the *same future timeline*, it is called **future value**.

Ex. Cash flow stream $(-2, 1, 1, 1)$ at $r = 10\%$ (yearly compounding, but any compounding convention works):

$$FV = -2(1.1)^3 + 1(1.1)^2 + 1(1.1) + 1 = 0.648$$

$$PV = -2 + \frac{1}{1.1} + \frac{1}{(1.1)^2} + \frac{1}{(1.1)^3} = 0.487$$

Check: $PV = FV/(1.1)^3 = 0.648/(1.1)^3 = 0.487$

General Formula for Present Value

Let $x = (x_0, x_1, x_2, \dots, x_n)$

$$\begin{aligned} PV(x) &= x_0 + \frac{x_1}{(1+r)} + \frac{x_2}{(1+r)^2} + \dots + \frac{x_n}{(1+r)^n} \\ &= \sum_{k=0}^n \frac{x_k}{(1+r)^k} \end{aligned}$$

Note that it's usually assumed that x_0 is negative, for the initial investment.

Main Theorem on Present Value

Two cash flow streams are equivalent for an ideal bank iff they have the same present value.

Internal Rate of Return

Captures the idea on the "return" on an investment.

The return on your investment is the return which would make the present value equal to 0.

Ex. Invest \$100 today, receive \$110 in one year. What is your return?

Intuitively: 10%, why?

$$x(-100, 110)$$
$$PV(x) = -100 + \frac{110}{1 + 0.1} = 0$$

Def. Let $x = (x_0, x_1, \dots, x_n)$ be a cash flow stream.

Then, the **internal rate of return** of this stream is the number r satisfying:

$$PV(x) = 0$$
$$x_0 + \frac{x_1}{1+r} + \frac{x_2}{(1+r)^2} + \dots + \frac{x_n}{(1+r)^n} = 0$$

To solve for this r , its more convenient to set

$$c = \frac{1}{(1+r)}$$

and solve

$$x_0 + x_1c + x_2c^2 + \dots + x_nc^n = 0$$

with the fundamental theory of algebra.

Ex. What is the internal rate of return of $(-2, 1, 1, 1)$?

Solve: $0 = -2 + c + c^2 + c^3$

$$c = 0.81 \quad \Rightarrow \quad c = \frac{1}{1+r} \quad \Rightarrow \quad r = 0.23$$

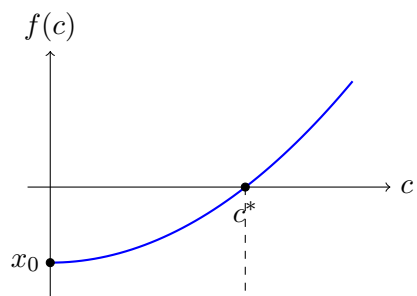
The polynomial can have many roots, which can be a problem. Fortunately, most standard investments have a well-defined IRR (see the theorem below).

Main Theorem on IRR

Suppose the cash flow stream $x = (x_0, x_1, \dots, x_n)$ has $x_0 < 0$ and $x_k \geq 0$ where at least one term is strictly positive. Then, there is a unique positive root to the equation

$$x_0 + x_1c + x_2c^2 + \dots + x_nc^n = 0$$

Proof: plot $f(c) = x_0 + x_1c + x_2c^2 + \dots + x_nc^n$



Since $f(0) = x_0 < 0$ and $f'(c) = x_1 + 2x_2c + \dots + nx_nc^{n-1} \geq 0$ for $c \geq 0$ (as each $x_k \geq 0$, with at least one strictly positive), f is nondecreasing and unbounded above. So f crosses zero exactly once, at some $c^* > 0$.

Net Present Value (NPV)

Computes all present values of *all* cash flows associated with an investment. Includes the investment term x_0 as well.

Ex. Given two cash flow vectors

(1) After 1 year: $(-1, 2)$

(2) After 2 years: $(-1, 0, 3)$

Using NPV solution (choose largest NPV):

$$-1 + \frac{2}{1.1} = 0.82 \quad \text{vs} \quad -1 + \frac{0}{1.1} + \frac{3}{1.1^2} = 1.48$$

NPV says choose option (2).

Using IRR (choose greatest IRR):

$$-1 + 2c = 0 \Rightarrow c = 0.5 \Rightarrow r = 1.0 \quad \text{vs} \quad -1 + 3c^2 = 0 \Rightarrow c = 0.577 \Rightarrow r = 0.7$$

IRR says choose option (1).

IRR and NPV give conflicting recommendations.

When scaled, IRR is scale-invariant, PV is not.

PV also distinguishes between single and repeated cash flows, IRR does not.

In general, NPV is preferred.

Ex. Scaling and repetition (PV at 10%):

Cash flows	$(-1, 2)$	$10 \cdot (-1, 2)$	$(-1, 2, -1, 2, \dots)$
PV	0.818	8.18	4.71
IRR	$r = 1.0$	$r = 1.0$	$r = 1.0$

NPV is preferred because it distinguishes between more cash flows, but this means the cash flows and assumptions need to be modelled carefully before the NPV analysis is done. E.g. the tree-cutting NPV analysis assumes it is done only *once*; if the process can be repeated many times, we arrive at a different answer. Methods of IRR analysis exist that make it agree with NPV (e.g. incremental IRR analysis).

Steps to Solving Cash Flow Problems

1. Write down the cash flow diagram for each alternative (include *all* cash flows).
2. Evaluate the NPV of each cash flow.
3. Choose the alternative with the largest NPV.

Cycle Problems

Relate to activities that can be repeated. With NPV you arrive at a different answer depending on how many times an activity is repeated, so the activities must be evaluated over the same time horizon. Two approaches:

1. Repeat both activities until they terminate at the same time.
2. Repeat both to infinity: for an activity with lifetime T and one-cycle present value PV_1 ,

$$PV_1 \sum_{k=0}^{\infty} \frac{1}{(1+r)^{Tk}} = PV_1 \cdot \frac{1}{1 - \frac{1}{(1+r)^T}}$$

Ex. Refrigerator Purchase

Model A: price \$600, yearly cost \$180 (beginning of year), life 3 years.

Model B: price \$800, yearly cost \$120 (beginning of year), life 5 years.

Interest rate is 10%. One cycle of each (price and first cost at time 0):

$$PV_A = 780 + 180 \sum_{k=1}^2 \frac{1}{1.1^k} = \$1,092.40 \quad PV_B = 920 + 120 \sum_{k=1}^4 \frac{1}{1.1^k} = \$1,300.38$$

Approach 1: equal horizons, 15 years = 5 cycles of A = 3 cycles of B

$$A : PV_A \left(1 + \frac{1}{1.1^3} + \frac{1}{1.1^6} + \frac{1}{1.1^9} + \frac{1}{1.1^{12}} \right) = \$3,341.11$$

$$B : PV_B \left(1 + \frac{1}{1.1^5} + \frac{1}{1.1^{10}} \right) = \$2,609.17$$

Approach 2: repeat to infinity

$$A : 1092.40 \cdot \frac{1}{1 - \frac{1}{1.1^3}} = \$4,392.70$$

$$B : 1300.38 \cdot \frac{1}{1 - \frac{1}{1.1^5}} = \$3,430.37$$

These are costs, so the smaller is better: choose Model B in both approaches.

Summing Geometric Series

$$\begin{aligned} S &= \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots \\ &= 1 + x(1 + x + x^2 + \cdots) = 1 + xS \end{aligned}$$

$$S(1 - x) = 1 \quad \Rightarrow \quad S = \frac{1}{1 - x} \quad (|x| < 1)$$

Lecture 3: Fixed Income Securities

Sep 21, 2026

Def. Financial Instruments

A legal obligation or claim having monetary value (eg. mortgage, stocks and bonds...)

Def. Security

A tradable financial instrument satisfying legal and regulatory requirements.

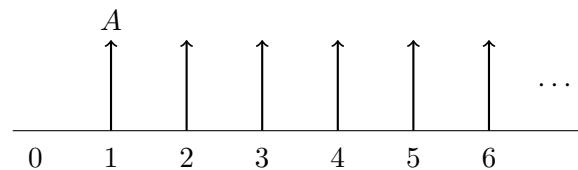
Def. Fixed Income Securities

Securities that promise a fixed income to the holder over some span of time. (eg. bonds, mortgages, annuities)

Annuity

A contract that pays the holder money periodically according to the fixed schedule.

Perpetual Annuity (Perpetuity) - Pays a fixed sum periodically, forever.



At a per period interest rate of r , the PV is

$$P = \sum_{k=1}^{\infty} \frac{A}{(1+r)^k} = \frac{A}{r}$$

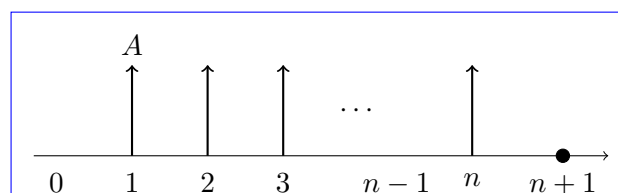
Note: Infinite sum!!

Ex. A perpetual annuity of \$1,000 each year at 10% interest/year has present value

$$P = \frac{A}{r} = \frac{\$1,000}{0.1} = \$10,000$$

Finite-Life Streams

A stream that pays A for n periods, starting at period 1. What is the PV?

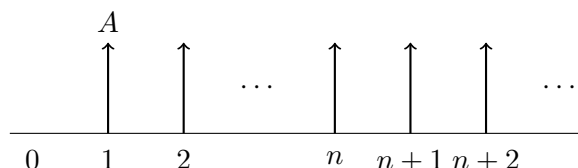


$$P = \sum_{k=1}^n \frac{A}{(1+r)^k} = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right)$$

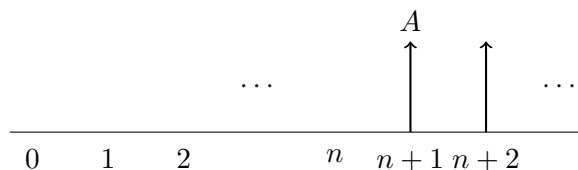
where r is the interest rate per period.

A finite life-stream can be thought as the difference between two perpetuities.

Perpetuity starting at period 1:



Perpetuity starting at period $n+1$:



$$PV = \frac{A}{r} - \frac{A}{r} \left(\frac{1}{(1+r)^n} \right)$$

$$PV_{fin} = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right)$$

Ex. Loan Calculation

Suppose \$1,000 is borrowed at 12% interest compounded monthly, repayed with equal monthly payments over 5 years. How much are the monthly payments?

Given: $P = \$1,000$ $r = \frac{12\%}{12} = 1\%/mo$ $n = 5 \times 12 = 60$

$$P = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right) \quad \Rightarrow \quad A = \frac{rP}{\left(1 - \frac{1}{(1+r)^n} \right)} = \$22.24/mo$$

Bonds

An agreement to pay money according to the rules of the issue.

Generally, bonds have

- Face Value, Par Value, Principal - an amount to be paid at the maturity date.
- Coupon Payments - an amount paid periodically, expressed as a % of the face value. Last coupon is paid at maturity.

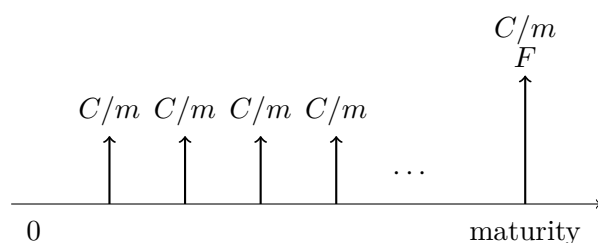
Ex. Given:

$$\text{Face Value} = \$1,000 \quad \text{Coupon} = 9\%$$

The yearly coupon amount is $\$1,000(0.09) = \90

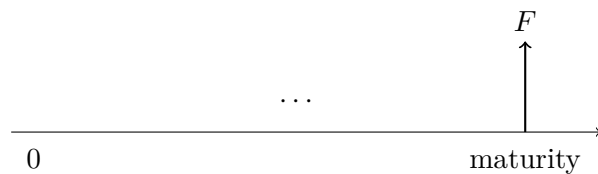
In general, coupons are paid m times per year, so each payment is C/m where C is the yearly coupon amount. In the U.S., coupons are usually paid every six months ($m = 2$), e.g. US treasury bonds with this example pay \$45 every 6 months ($\$90/2$).

Payoff Structure of a Bond



where C is the annual coupon amount, and F is the face value.

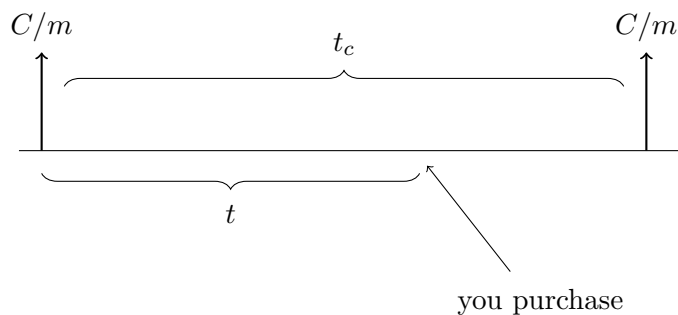
When there are no coupons, the only payoff is the face value at maturity.



This is known as a **zero-coupon bond**, or a zero.

Accrued Interest

Ignored by bond price quotes, but must be added to the price.



You must pay the previous owner their “portion” of the next coupon payment.

$$AI = \frac{\text{days since last coupon}}{\text{days in current coupon period}} \times (C/m) = \frac{t}{t_c}(C/m)$$

Ex. A treasury bond is purchased on May 8 and matures on August 15 in some future year. The coupon rate is 9% and coupons are paid every Feb 15 and Aug 15. How much accrued interest is owed?

On May 8th, 83 days have passed since the last coupon and 99 days remain until the next. With $C/m = 9\%/2 = 4.5\%$:

$$AI = \frac{83}{83 + 99} \times 4.5\% = 2.05\%$$

Def. Clean price: the market bond price (quoted, doesn’t include accrued interest).

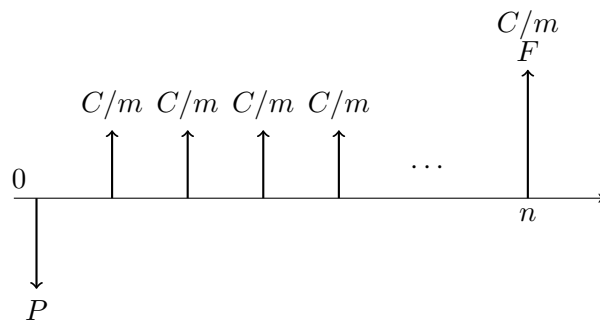
Dirty (cash) price: the clean price plus accrued interest, i.e. what you actually pay.

Government securities:

- Bill: up to 1 year
- Note: 1 to 10 years
- Bond: 10 to 30 years

Bond Price and Yield

The whole cash flow stream of purchasing a bond can be mapped out, where P is the entire amount paid for the bond: the market price plus accrued interest (the dirty price).



The IRR of this cash flow stream is the **yield** (or **yield to maturity**) of the bond, denoted λ . It captures the rate of return from buying the bond and holding it to maturity.

$$P = \underbrace{\frac{F}{\left(1 + \frac{\lambda}{m}\right)^n}}_{\text{PV of face}} + \underbrace{\frac{C/m}{\lambda/m} \left[1 - \frac{1}{\left(1 + \frac{\lambda}{m}\right)^n} \right]}_{\text{annuity formula}}$$

- P : price including accrued interest
- F : face value
- m : coupon payments per year (compounding m times per year),
- C/m : amount of each coupon payment
- n : amount of coupon payments remaining.
- λ/m : yield per coupon period, n : coupon payments left

Note this equation only holds when λ is the yield.

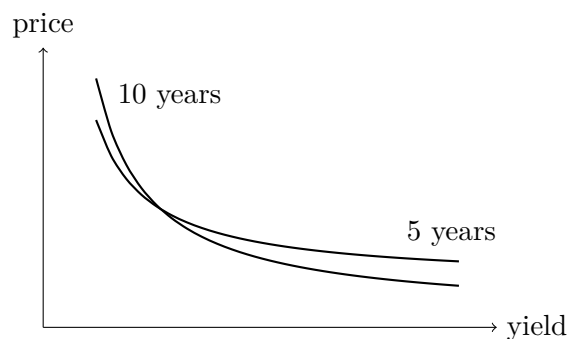
Ex. Price of a bond with par value \$1,000 paid in ten years, coupon rate 10%, yield 12%, coupons paid semi-annually. Then $n = 20$, $C/m = 50$, $\lambda/m = 6\%$:

$$P = 50 \cdot \frac{1 - \frac{1}{(1.06)^{20}}}{0.06} + \frac{1000}{(1.06)^{20}} = 50(11.47) + 311.80 \approx \$885.30$$

The Price-Yield Curve

For a fixed maturity, price and yield are inversely related

The longer the time to maturity, the more sensitive the price of the bond is to the yield.



The **current yield** is the coupon divided by the current price. In this course, **yield** always means **yield to maturity**.

Bond Ratings

Standard & Poor's and Moody's are two independent bond rating agencies.

	S&P	Moody's	Meaning
Investment	AAA	Aaa	Best quality, smallest credit risk (US gov. bonds)
Grade	AA	Aa	High grade
	A	A	High to medium grade
	BBB	Baa	Medium grade
Speculative	BB	Ba	Judged to be speculative
Grade	B	B	Increasingly speculative
(Junk Bonds)	CCC	Caa	Danger of default
	CC	Ca	Very speculative, high chance of default
	C	C	Small chance of not defaulting
	D	D	In default

Lecture 4: